

Assignment 2

DAX Calculations

Calculated Columns

- **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.
- **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.
- **Create a Calculated Column to Categorize Sales:** Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

The screenshot displays the Microsoft Power BI Desktop interface. At the top, the formula bar contains the DAX expression: `Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`. Below the formula bar, the main workspace area shows a prompt to "Build visuals with your data" and a visual selection pane. On the right side, the "Data" pane is open, showing a list of fields from the "Order Details" table. The fields listed include City, CustomerName, Order Date, Order ID, State, Amount, Category, Order ID, Profit, Quantity, Revenue, Sales Category, Sub-Category, Sales target, and Month of Ord... The "Category Type" field, which is the calculated column being created, is highlighted with a red box. A tooltip is visible over the "Category Type" field, displaying its name: "Name 'Order Details'[Category Type]".

Power BI Desktop interface showing a calculated column formula and the Data pane.

Structure: Name: Revenue, Data type: Whole number

Formatting: Format: Whole number, Summarization: Sum, Data category: Uncategorized

Properties: Sort by column, Data groups, Manage relationships, New column

Formula Bar: 1 Revenue = 'Order Details'[Amount] * 'Order Details'[Quantity]

Visualizations: Build visual, Add data fields here, Filters on this page, Filters on all pages, Values, Drill through, Cross-report, Keep all filters, Add drill-through fields here

Data: List of Orders, Order Date, Order ID, State, Order Details, Amount, Category, Category Type, Order ID, Profit, Quantity, Revenue, Sales Category, Sub-Category, Sales target, Category, Month of Ord...

Power BI Desktop interface showing a calculated column formula and the Data pane.

Structure: Name: Sales Category, Data type: Text

Formatting: Format: Text, Summarization: Don't summarize, Data category: Uncategorized

Properties: Sort by column, Data groups, Manage relationships, New column

Formula Bar: 1 Sales Category =
2 IF(
3 'Order Details'[Amount] > CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details')),
4 "Above Average",
5 "Below Average"
6)

Visualizations: Build visual, Add data fields here, Filters on this page, Filters on all pages, Values, Drill through, Cross-report, Keep all filters, Add drill-through fields here

Data: List of Orders, Order Date, Order ID, State, Order Details, Amount, Category, Category Type, Order ID, Profit, Quantity, Revenue, Sales Category, Sub-Category, Sales target, Category, Month of Ord...

1 . Category Type

- **Formula:** `'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`
- **Observation:** "We combined the Category and Sub-Category into a single column (e.g., *Furniture - Chairs*). This makes it much easier to read and filter product hierarchies in a single view without needing multiple slicers."

2 . Revenue per Order

- **Formula:** `'Order Details'[Amount] * 'Order Details'[Quantity]`
- **Observation:** "We calculated the total money earned per line item by multiplying unit amount by quantity sold. This helps the business track the actual sales revenue generated by each order."

3 . Sales Category

- **Formula:** `IF('Order Details'[Amount] > CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details'))), "Above Average", "Below Average")`
- **Observation:** "We classified each transaction as 'Above Average' or 'Below Average' by comparing its amount to the overall average sales value. This allows us to quickly highlight high-value sales orders for performance analysis."

Calculated Measures

1. **Order Count-** Create a measure that calculates the total number of orders in the Order Details table.
2. **Average Profit in Delhi-** Create a measure that calculates the average profit for orders placed in Delhi.
3. **Year-to-Date (YTD) Sales-** Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

Power BI Desktop interface showing the formula bar and the Data pane.

Name: Total Orders
Format: Whole number
Data category: Uncategorized

Formula Bar: `Total Orders = DISTINCTCOUNT('Order Details'[Order ID])`

Data Pane:

- List of Orders
 - ☐ Avg Profit C
 - ☐ City
 - ☐ CustomerN
 - ☐ Order Date
 - ☐ Order ID
 - ☐ State
 - ☒ Total Order
 - ☐ YTD Sales
- Order Details
 - ☐ Amount
 - ☐ Category
 - ☐ Category Ty
 - ☐ Order ID
 - ☐ Profit
 - ☐ Quantity
 - ☐ Revenue
 - ☐ Sales Categ
 - ☐ Sub-Cateo

Power BI Desktop interface showing the formula bar and the Data pane.

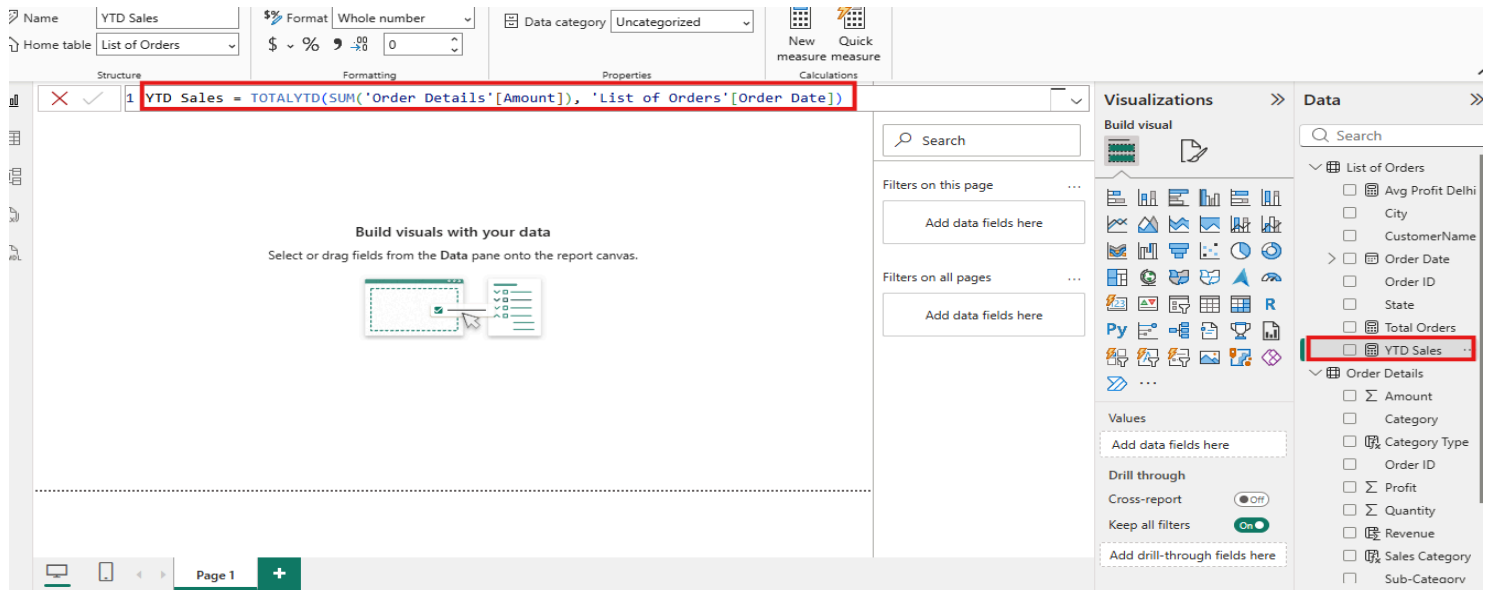
Name: Avg Profit Delhi
Format: General
Data category: Uncategorized

Formula Bar:

```
1 Avg Profit Delhi =  
2 CALCULATE(  
3     AVERAGE('Order Details'[Profit]),  
4     'List of Orders'[City] = "Delhi"  
5 )
```

Data Pane:

- List of Orders
 - ☒ Avg Profit ...
 - ☐ City
 - ☐ CustomerName
 - ☐ Order Date
 - ☐ Order ID
 - ☐ State
 - ☐ Total Orders
 - ☐ YTD Sales
- Order Details
 - ☐ Amount
 - ☐ Category
 - ☐ Category Type
 - ☐ Order ID
 - ☐ Profit
 - ☐ Quantity
 - ☐ Revenue
 - ☐ Sales Category
 - ☐ Sub-Category



Total Orders

- **Formula:** `Total Orders = DISTINCTCOUNT('Order Details'[Order ID])`
- **Observation:** Tracks the exact number of unique customer orders placed, measuring overall transaction volume.

Average Profit in Delhi

Formula: `Avg Profit Delhi = CALCULATE AVERAGE('Order Details'[Profit]), 'List of Orders'[City] = "Delhi")`

Observation: Evaluates regional profitability specifically for Delhi to analyze performance in key market locations.

Year-to-Date (YTD) Sales

- **Formula:** `YTD Sales = TOTALYTD(SUM('Order Details'[Amount]), 'List of Orders'[Order Date])`
- **Observation:** Accumulates total sales revenue from the beginning of the calendar year to track progress toward annual revenue goals.

Data Visualization

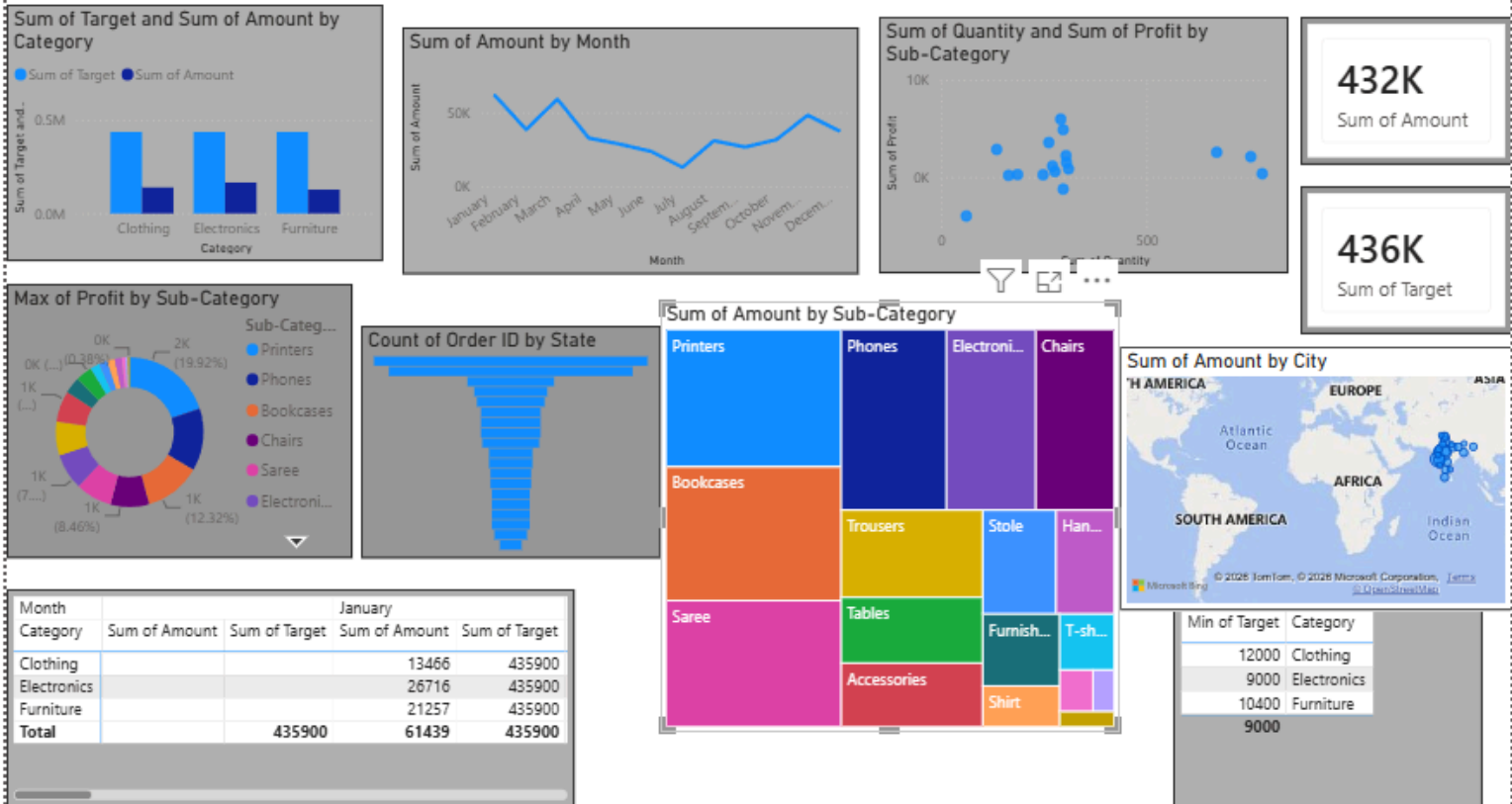
Create the following visualizations using Power BI:

- **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.
- **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.
- **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.
- **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.
- **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison

and analysis. Also, create a multi-row card to display the minimum target for each segment.

- **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.
- **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.
- **Sales Distribution by Sub-Category:** Represent the sales distribution across different
- sub-categories using a treemap.
- **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.

Dashboard



1. Sales Target Achievement by Category (Clustered Column Chart)

- **Observation:** Compares actual sales revenue against set sales targets across Clothing, Electronics, and Furniture categories.
- **Use Case:** Identifies which product categories met or missed their monthly financial performance goals.

2. Max Profit Margin by Sub-Category (Donut Chart)

- **Observation:** Displays the maximum profit margin achieved by each individual product sub-category.

- **Use Case:** Highlights top-performing product sub-categories to optimize pricing and inventory focus.

3. Monthly Sales Trend (Line Chart)

- **Observation:** Illustrates the month-over-month sales revenue movement over time.
- **Use Case:** Helps forecast demand, track seasonality, and time strategic marketing campaigns.

4. Comparison of Profit and Quantity by Sub-Category (Scatter Chart)

- **Observation:** Examines the relationship between total quantity sold and profit generated across sub-categories.
- **Use Case:** Separates high-margin niche items from high-volume drivers for better inventory planning.

5. Comparison of Total Sales Amount and Target (Cards & Multi-row Card)

- **Observation:** Shows high-level KPIs for overall sales (432K), overall target (436K), and minimum targets by segment.
- **Use Case:** Offers executives an instant 2-second snapshot of overall business target compliance.

6. Sales Performance Matrix (Matrix Visual)

- **Observation:** Breaks down actual sales versus targets in a detailed grid across categories and months.
- **Use Case:** Serves as a primary reference for granular financial reporting, auditing, and month-by-month reconciliation.

7. Geographic Sales Analysis (Map Visual)

- **Observation:** Maps total sales revenue geographically across different cities and regions.
- **Use Case:** Pinpoints high-performing sales locations to guide regional resource and staff allocation.

8. Sales Distribution by Sub-Category (Treemap)

- **Observation:** Uses proportional visual blocks to show revenue contribution by sub-category.
- **Use Case:** Delivers a quick visual breakdown of market share distribution among product lines.

9. Order Count Analysis by State (Funnel Chart)

- **Observation:** Ranks states by total order volume from highest to lowest in a funnel flow.
- **Use Case:** Uncovers key geographic markets to prioritize supply chain and targeted advertising investments.